# 2016-Summer Paper 2 – Urology

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## Q1. CA bladder, etiology, newer investigations and treatment.

**Type:** LAQ (25 marks)

### Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

### Aetiology (risk factors)

| Factor | Key facts |
|---|---|
| **Smoking** | 30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting |
| **Occupation** (5–10%) | Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps) |
| Environment | **Arsenic** in water (West Bengal, Bangladesh, Taiwan) |
| Radiation | Pelvic radiotherapy 1.5–4 times; latency 15–30 years |
| Drugs | **Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting) |
| Chronic inflammation | *Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma |
| Genetic | First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion |

- **Also know:** obesity raises risk; diet has no proven role.

### Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

| TNM (tumour, node, metastasis), 2017 | Extent |
|---|---|
| Ta / Tis | Papillary non-invasive / flat CIS |
| T1 | Lamina propria |
| T2a / T2b | Inner / outer half of muscle |

| TNM (tumour, node, metastasis), 2017 | Extent |
|---|---|
| T3a / T3b | Perivesical fat, microscopic / macroscopic |
| T4a / T4b | Prostatic stroma, uterus, vagina / pelvic or abdominal wall |
| N1 / N2 / N3 | One pelvic node / several / common iliac |

## Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.
- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris System** reporting).

## Newer investigations

| Test | Principle | Value |
|---|---|---|
| **Blue-light cystoscopy (PDD, photodynamic diagnosis)** with HAL (hexaminolevulinate) | Porphyrins in tumour fluoresce red under blue light | About one-third more CIS; recurrence benefit inconsistent |
| **NBI (narrow-band imaging)** | 415 and 540 nm light absorbed by haemoglobin highlights vessels | More tumours; many false positives |
| **Urine biomarkers** | NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder | More sensitive, less specific than cytology; **none replaces cystoscopy** |
| **mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)** | Muscle-line integrity scored 1–5 | **≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT |
| **En bloc TURBT (transurethral resection of bladder tumour)** | Tumour removed in one piece with detrusor | Better staging and muscle sampling, fewer perforations; recurrence similar |

- **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases.

## Treatment

**1. TURBT**: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

### 2. NMIBC by risk (AUA (American Urological Association))

| Risk | Definition | Treatment |
|---|---|---|
| Low | Solitary low-grade Ta ≤3 cm; PUNLMP | Single instillation; surveillance |
| Intermediate | Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm | Chemotherapy or BCG; maintenance 1 year |
| High | High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion | **BCG induction + 3-year maintenance**; early cystectomy if very high risk |

- **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression.
- **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200.

**3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0**

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.
- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

**4. Metastatic**: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

## Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

**Diagram to draw:** bladder wall layers with Ta, Tis, T1–T4 marked.

> **Key points to remember:**
>
> - Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
> - Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
> - High-risk NMIBC: BCG with 3-year maintenance.
> - MIBC: cisplatin NAC + cystectomy; trimodality in selected.

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## Q2. Describe etiology, clinical presentation and management of urethral strictures.

**Type:** LAQ (25 marks)

### Definition

- A fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing mucosa. It is an epithelial scar with **spongiofibrosis**.
- Posterior narrowing is named differently: after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses three-quarters of the lumen, so symptoms appear late.

### Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries and retrograde flow from the glans) allows the bulb to be divided and mobilised.

### Aetiology

| Category | Examples |
|---|---|
| **Idiopathic** (commonest, with iatrogenic) | No cause found |
| **Iatrogenic** | Catheters (common in LMICs (low- and middle-income countries)), TURP (transurethral resection of prostate), bladder tumour resection, cystoscopy, hypospadias repair, radiotherapy |
| **Traumatic** | **Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI) |
| **Inflammatory** | Gonorrhoea (now rare); **lichen sclerosus (LS)** |
| Congenital | Rarest; infants only |

- **LS**: chronic skin disease starting at the glans and meatus. Urine forced into Littre glands causes inflammation and deep spongiofibrosis. It recurs in any skin used to repair it.
- **Jordan's spectrum**: mucosal fold → iris → full-thickness → deep spongiofibrosis → periurethral fibrosis → fistula or abscess.

### Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- Retention after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

### Evaluation (define 4 things: location, length, depth, density)

| Test | Role |
|---|---|
| **Uroflowmetry and PVR (post-void residual)** | Low, plateau-shaped flow; screen and follow-up |
| **Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)** | Site and length; steep oblique views; done by the surgeon |
| **Flexible urethroscopy** | Calibre, appearance; paediatric scope sees beyond |
| **Sonourethrography** | Length and depth of spongiofibrosis |

| Test | Role |
|---|---|
| MRI (magnetic resonance imaging) | PFUI defect alignment; not routine |

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation. In retention, place a **suprapubic catheter** rather than force a dilation.

## Management

First agree the aim: **palliation or cure**. The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned, because repeated endoscopy makes later urethroplasty harder.

**1. Dilation**: soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Curative only for epithelial strictures without spongiofibrosis.

**2. DVIU (direct vision internal urethrotomy)**

- Heals by **secondary intention**: a race between epithelialisation and wound contraction.
- Best for **bulbar strictures <1.5 cm**, without dense spongiofibrosis, at first attempt. A second urethrotomy adds almost nothing. Laser gives no advantage over cold knife.
- Complications: recurrence (commonest), bleeding, extravasation, ED (erectile dysfunction) from deep cuts.
- Adjuncts: **intermittent self-dilation**; mitomycin C injection. **Optilume** paclitaxel-coated balloon (**ROBUST III**: 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy).

**3. Urethroplasty**

| Technique | Indication and key points |
|---|---|
| **Excision and primary anastomosis (EPA)** | **Gold standard**, about 90–95% success; short bulbar strictures (1–2 cm, up to 3–5 cm proximally). Excise all scar, spatulate widely, tension-free. **Non-transecting** and **vessel-sparing** variants preserve bulbar arteries |
| **BMG (buccal mucosal graft) onlay** | Longer bulbar strictures: **dorsal onlay (Barbagli)**, ventral onlay, **dorsal inlay (Asopa)**. Graft equals flap. Kulkarni one-sided repair for panurethral strictures |
| **Augmented anastomosis** | Excise the narrowest part, anastomose the floor, graft the roof |
| **Penile skin flaps** (Orandi, Quartey, circular) | Penile strictures; skin island carried on dartos |
| **Staged repair** (Johanson) | Stage 1 open and graft; tubularise after 6 months or more. For **LS**, failed hypospadias, infection |
| **Perineal urethrostomy** | Complex or panurethral disease, elderly or after failures |

- **Onlay is better than tube** repairs; **avoid genital skin in LS** (use oral mucosa). Topical clobetasol for early LS meatal disease.
- Meatal stenosis: meatotomy; fossa navicularis: flap or graft.
- **PFUI**: suprapubic catheter, then delayed **perineal anastomotic urethroplasty** after 3 months or more. Progressive steps: bulbar mobilisation, intracrural (corporal) separation, inferior pubectomy, rerouting of the urethra.
- **Urethral stents** (UroLume) are withdrawn; AUA (American Urological Association): no stent if urethroplasty is possible.

## Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

## Follow-up

- Uroflow, PVR and symptom score; RUG or cystoscopy if flow falls. Most EPA recurrences occur in the first 6 months; graft results decline with time.

## Recent advances

- Tissue-engineered oral mucosa; robotic reconstruction of posterior stenosis.

**Diagram to draw:** RUG of a bulbar stricture, and cross-section of dorsal onlay versus ventral onlay.

> **Key points to remember:**
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Assess location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
> - DVIU only for bulbar <1.5 cm at first attempt; do not repeat.
> - EPA is gold standard; BMG onlay for longer; staged repair for LS.

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# Q3a. Trans Uretero Ureteric anastomoses

**Type:** SAQ (10 marks)

## Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

## Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

## Contraindications

| Absolute | Relative (disease that may involve both ureters) |
|---|---|
| Donor ureter too short to reach without tension | Stone disease, **urothelial carcinoma** |
| Diseased recipient ureter | Retroperitoneal fibrosis, pelvic radiation |
| | Chronic pyelonephritis, genitourinary tuberculosis |

- Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk).

## Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

## Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

## Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

**Diagram to draw:** donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

> **Key points to remember:**
>
> - End-to-side, spatulated, tension-free; cross above the IMA.
> - Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
> - Risk: unilateral injury becomes bilateral.

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# Q3b. Acquired renal cystic disease

**Type:** SAQ (10 marks)

## Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

## Epidemiology

| Duration of renal failure | Prevalence |
|---|---|
| Before dialysis | About 10% |
| 3 years of dialysis | 44% |
| 5 years | 60% |
| ≥10 years | >90% |

- Male:female 3:1; independent of the cause of renal failure.

## Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

## Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate crystals, good prognosis); papillary RCC is also common.

## Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

| Feature | ARCD | ADPKD (autosomal dominant polycystic kidney disease) |
|---|---|---|
| Kidney size | Small (<300 g) | Large (>800 g) |
| Cysts | Small | Large |
| Extrarenal | None | Liver cysts, aneurysms |

## Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.

- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

**Diagram to draw:** small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

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# Q3c. Bladder diverticulum.

**Type:** SAQ (10 marks)

## Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

## Classification

| Feature | Congenital (primary) | Acquired (secondary) |
|---|---|---|
| Age | Children (<10 years) | Men >60 years |
| Cause | Weak detrusor at the ureteric hiatus | **BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder |
| Number, site | Solitary; **paraureteral (Hutch)** | Often multiple, anywhere |
| Bladder wall | Smooth | Trabeculated |
| Associations | VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes | Stones, cancer |

- **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum.

## Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.
- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

## Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

## Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.
- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.

- **Tumour inside**: partial or radical cystectomy.

## Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.
- Complications: ureteric injury, leak.

**Diagram to draw:** paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

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## Q3d. Botulinim neurotoxin.

**Type:** SAQ (10 marks)

*Assumed meaning: botulinum neurotoxin (spelling error in the question).*

### Introduction

- Toxin of *Clostridium botulinum*; serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**; type B is short-acting.
- 150 kDa: heavy chain (100 kDa) plus zinc-dependent light chain (50 kDa).

### Mechanism

1. Heavy chain binds the cholinergic nerve terminal; the toxin is internalised.
2. Light chain enters the cytosol.
3. It **cleaves SNAP-25 (synaptosomal-associated protein 25)**, a SNARE (vesicle-docking) protein, so vesicles cannot fuse and acetylcholine release stops.

- **Afferent action**: less ATP (adenosine triphosphate) and substance P release; fewer TRPV1 and P2X3 receptors, so urgency falls too.
- Effect wears off as nerve terminals sprout (about 6–9 months).

### Urological uses

| Indication | Dose and evidence |
|---|---|
| **Idiopathic OAB (overactive bladder)**, third line | **100 U**; incontinence down about 59%, about 25% dry. ABC trial: equal to anticholinergic, more complete dryness. ROSETTA: 200 U equal to SNM (sacral neuromodulation), more UTI (urinary tract infection) |
| **NDO (neurogenic detrusor overactivity)**: SCI (spinal cord injury), MS (multiple sclerosis) | **200 U** (DIGNITY trials): lower detrusor pressure, larger capacity |
| Children with NDO | About 10 U/kg (maximum 360 U) |
| **DSD (detrusor–sphincter dyssynergia)** | Intrasphincteric injection in men unable to do CISC (clean intermittent self-catheterisation) |
| Others | Bladder pain syndrome; prostate injection for benign enlargement failed versus sham |

### Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U) or 30 sites (200 U)**, sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

### Side effects

- **Retention needing CISC**: 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; higher in NDO. Each 10 mL rise in baseline PVR raises risk 27%. Patients must accept CISC beforehand.
- **UTI** about 20%; haematuria; rarely generalised weakness.
- Contraindications: active UTI, myasthenia gravis, unwillingness to self-catheterise.
- Repeat injections when the effect fades; response is maintained.

**Diagram to draw:** cholinergic synapse showing SNAP-25 cleavage.

**Key points to remember:**

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Main risks: retention needing CISC, UTI.

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## Q3e. Catheter free trial

**Type:** SAQ (10 marks)

*Assumed meaning: trial without catheter (TWOC) after acute urinary retention.*

### Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

### Types of AUR

| Spontaneous | Precipitated |
|---|---|
| BPH (benign prostatic hyperplasia) progression | Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs |
| TWOC fails more often; about 75% later need surgery | Better TWOC outcome; about a quarter need surgery |

### Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

### Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

### Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

### Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

### Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

### After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.
- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

**Diagram to draw:** algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

**Key points to remember:**

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

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## Q3f. Urethral caruncle.

**Type:** SAQ (10 marks)

### Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

### Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

### Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

### Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

### Differential diagnosis

| Lesion | Distinguishing feature |
|---|---|
| **Urethral mucosal prolapse** | **Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women |
| **Urethral carcinoma** | Firm, ulcerated, indurated, fixed; bleeding mass; biopsy |
| **Urethral diverticulum** | Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging) |
| Skene gland cyst or abscess | Lateral to meatus; distorts it; does not communicate |
| Others | Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus |

- Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied.

### Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

### Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions:

1. Grasp the tip and apply traction to expose the base.
2. Excise at the base.
3. Reapproximate mucosa with absorbable sutures if the defect is large.
4. Foley catheter for a few days.

- Complications: bleeding, recurrence, meatal stenosis.

**Diagram to draw:** meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

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# Rapid Revision Sheet – 2016-Summer Paper 2

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. CA bladder, etiology, newer investigations and treatment. (LAQ)**

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

**Q2. Describe etiology, clinical presentation and management of urethral strictures. (LAQ)**

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Assess location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU only for bulbar <1.5 cm at first attempt; do not repeat.
- EPA is gold standard; BMG onlay for longer; staged repair for LS.

**Q3a. Trans Uretero Ureteric anastomoses (SAQ)**

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
- Risk: unilateral injury becomes bilateral.

**Q3b. Acquired renal cystic disease (SAQ)**

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

**Q3c. Bladder diverticulum. (SAQ)**

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

**Q3d. Botulinim neurotoxin. (SAQ)**

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Main risks: retention needing CISC, UTI.

**Q3e. Catheter free trial (SAQ)**

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

**Q3f. Urethral caruncle. (SAQ)**

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.